



Does going to university in a different country affect your mental health? A Japanese international university surveyed its students in 2018 and published a study the following year that was approved by several ethical and regulatory boards.

The study found that international students have a higher risk of mental health difficulties than the general population, and that social connectedness (belonging to a social group) and acculturative stress (stress associated with joining a new culture) are predictive of depression.

Explore the `students` data using PostgreSQL to find out if you would come to a similar conclusion for international students and see if the length of stay is a contributing factor.

Here is a data description of the columns you may find helpful.

Field Name	Description
<code>inter_dom</code>	Types of students (international or domestic)
<code>japanese_cate</code>	Japanese language proficiency
<code>english_cate</code>	English language proficiency
<code>academic</code>	Current academic level (undergraduate or graduate)
<code>age</code>	Current age of student
<code>stay</code>	Current length of stay in years
<code>todep</code>	Total score of depression (PHQ-9 test)
<code>tosc</code>	Total score of social connectedness (SCS test)
<code>toas</code>	Total score of acculturative stress (ASISS test)

Projects Data DataFrame as df

```
SELECT stay,
        COUNT(*) AS count_int,
        ROUND(AVG(todep), 2) AS average_phq,
        ROUND(AVG(tosc), 2) AS average_scs,
        ROUND(AVG(toas), 2) AS average_as
FROM students
WHERE inter_dom = 'Inter'
GROUP BY stay
ORDER BY stay DESC
```

index	...	↑↓	stay	...	↑↓	count_int	...	↑↓	average_phq	...	↑↓	average_scs	...	↑↓	average_as
			0			10			1			13			
			1			8			1			10			
			2			7			1			4			
			3			6			3			6			
			4			5			1			0			
			5			4			14			8.57			
			6			3			46			9.09			
			7			2			39			8.28			
			8			1			95			7.48			

Rows: 9

[Expand](#)

Projects Data DataFrame as df1

```
-- Impacto da conectividade social e depresao , extra ao projeto inicial
SELECT
CASE
    WHEN tosc <= 24 THEN 'Baixa conectividade social (0-24)'
    WHEN tosc BETWEEN 25 AND 36 THEN 'Media conectividade social (25-36)'
    ELSE 'Alta conectividade social +37'
END AS conectividade_social,
COUNT(*) AS Estudantes,
ROUND(AVG(todep),2) Depressao_Media
FROM students
WHERE inter_dom = 'Inter'
GROUP BY conectividade_social
ORDER BY conectividade_social DESC;
```

index	...	↑↓	conectividade_social	...	↑↓	estudantes
			0			Media conectividade social (25-36)
			1			Baixa conectividade social (0-24)
			2			Alta conectividade social +37

Rows: 3

[Expand](#)

Projects Data DataFrame as df2

```
-- Impacto do genero e religiao extra ao projeto inicial
SELECT gender,
        religion,
        COUNT(*) AS student_count,
        ROUND(AVG(todep), 2) AS average_phq,
        ROUND(AVG(tosc), 2) AS average_scs,
        ROUND(AVG(toas), 2) AS average_as
FROM students
WHERE inter_dom = 'Inter'
GROUP BY gender, religion
ORDER BY gender, average_phq DESC
```

index	...	↑↓	gender	...	↑↓	religion	...	↑↓	student_count	...	↑↓	average_phq
			0			Female			No			86
			1			Female			Yes			42
			2			Male			No			40
			3			Male			Yes			33

Rows: 4

[Expand](#)

Projects Data DataFrame as df3

```
-- impacto de ser estudante internacional ou nativo ,extra ao projeto inicial
SELECT inter_dom,
       COUNT(*) AS total_students,
       ROUND(AVG(todep), 2) AS average_phq,
       ROUND(AVG(tosc), 2) AS average_scs,
       ROUND(AVG(toas), 2) AS average_as
FROM students
WHERE inter_dom IN ('Dom', 'Inter') -- a base de dados tem espaços em branco em vez de null
GROUP BY inter_dom;
```

...	↑↓	i...	↑↓	total_st...	...	↑↓	av...	...	↑↓	av...	...	↑↓	a...	...	↑↓
0		Inter		201			8.04			37.42			75.56		
1		Dom		67			8.61			37.64			62.84		

Rows: 2

[Expand](#)

Projects Data DataFrame as d

```
-- impacto de falar a linguagem fluientemente, extra ao projeto inicial
SELECT japanese_cate,
       COUNT(*) AS student_count,
       ROUND(AVG(todep), 2) AS average_phq,
       ROUND(AVG(tosc), 2) AS average_scs,
       ROUND(AVG(toas), 2) AS average_as
FROM students
WHERE inter_dom = 'Inter'
GROUP BY japanese_cate
ORDER BY average_phq
```

...	↑↓	japane...	...	↑↓	studen...	...	↑↓	av...	...	↑↓	av...	...	↑↓	a...	...	↑↓
0		High			25			7.4			38.48			72		
1		Low			91			7.91			36.99			76.65		
2		Average			85			8.38			37.56			75.45		

Rows: 3

[Expand](#)

Projects Data DataFrame as d

```
-- impacto da idade , extra ao projeto inicial
SELECT age,
       COUNT(*) AS student_count,
       ROUND(AVG(todep), 2) AS average_phq,
       ROUND(AVG(tosc), 2) AS average_scs,
       ROUND(AVG(toas), 2) AS average_as
FROM students
WHERE inter_dom = 'Inter'
GROUP BY age
ORDER BY age DESC
```

...	↑↓	...	↑↓	studen...	...	↑↓	av...	...	↑↓	av...	...	↑↓	a...	...	↑↓
0		31		4			5.75			43.5			80.25		
1		30		3			9.33			41			97.33		
2		29		4			3.75			43			63.5		
3		28		3			3.33			39			71		
4		27		1			10			35			42		
5		25		9			6.11			37.33			80.78		
6		24		6			4.67			42.33			74.33		
7		23		12			9.67			32			81.25		
8		22		14			8.36			38.14			70.43		
9		21		39			9.23			37.74			75.23		
10		20		34			7.35			38.21			73.26		
11		19		41			8.44			37.9			74.1		
12		18		28			8.75			34.11			80.61		
13		17		3			4.67			37.33			70.67		

Rows: 14

[Expand](#)